
Problem 8.21.

Part a. Let $ADD = \{\langle x, y, z\rangle \mid x, y, z > 0$ are binary integers and $x + y = z\}$.
Show that $ADD \in$ L.

Proof. □

Part b. Let $PAL - ADD = \{\langle x, y\rangle \mid x, y > 0$ are binary integers where $x + y$ is an integer whose binary representation is a palindrome$\}$. (Note that the binary representation of the sum is assumed not to have leading zeros. A palindrome is a string that equals its reverse.)
Show that $PAL - ADD \in$ L.

Proof. □